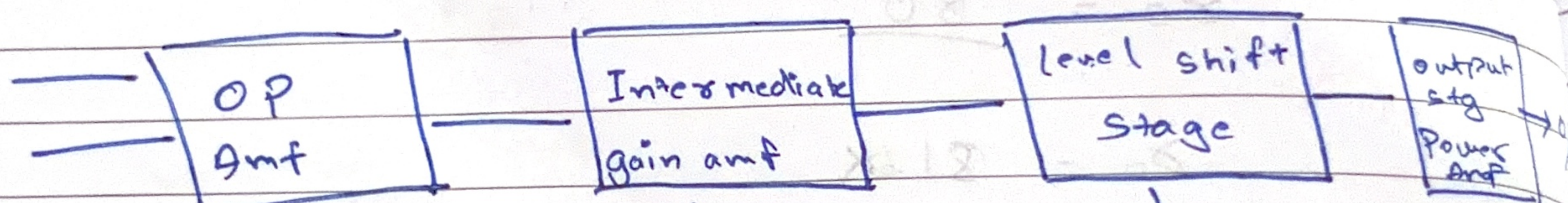
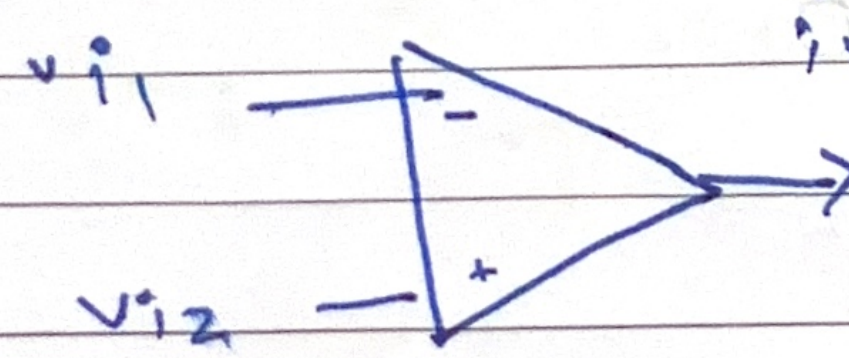
Module-3

Op-ampf → is a high gain, high input impedance and no O/P impedance and integrated circuit DC ampf.
it comprises of 4 stages
i) Diff ampf
ii) gain
iii) level shifter stage
iv) Output power ampf.





$V_o = A_v(v_{i_2} - v_{i_1})$

Features of Op Ampf

i) inf. differential voltage gain $A_{od} = \frac{V_o}{V_i} = \frac{V_o}{V_{i_2} - V_{i_1}} = \infty$ $V_{i_2} \neq V_{i_1}$

ii) inf. i/p impedance, $I_1 = I_2 = 0$, $R_{in} = \infty$

iii) 0 o/p impedance, O/P is directly taken from the load it is independent of the load.

iv) 0 o/p Voltage, when I/P V is Zero

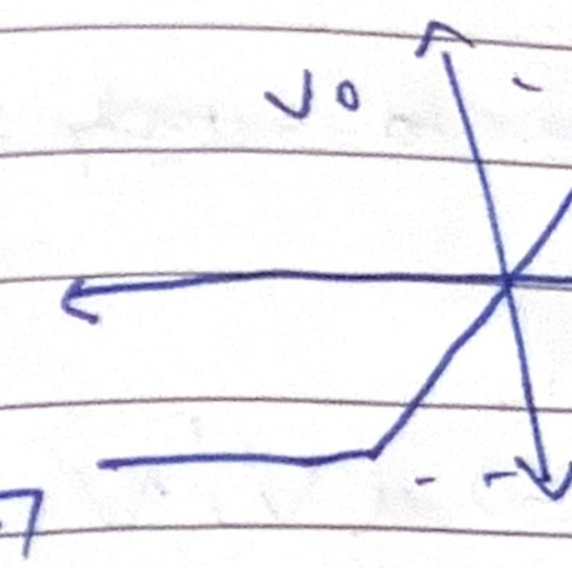
v) inf CMRR (common mode rejection ratio)

vi) inf. slew rate

vii) 0 SVRR

Op amp works i

open loop



Op-Amp ope

Practical

i) O/P of voltage I/P sign mV.

ii) I/P def btw to

iii) I/P I=

iv) I/

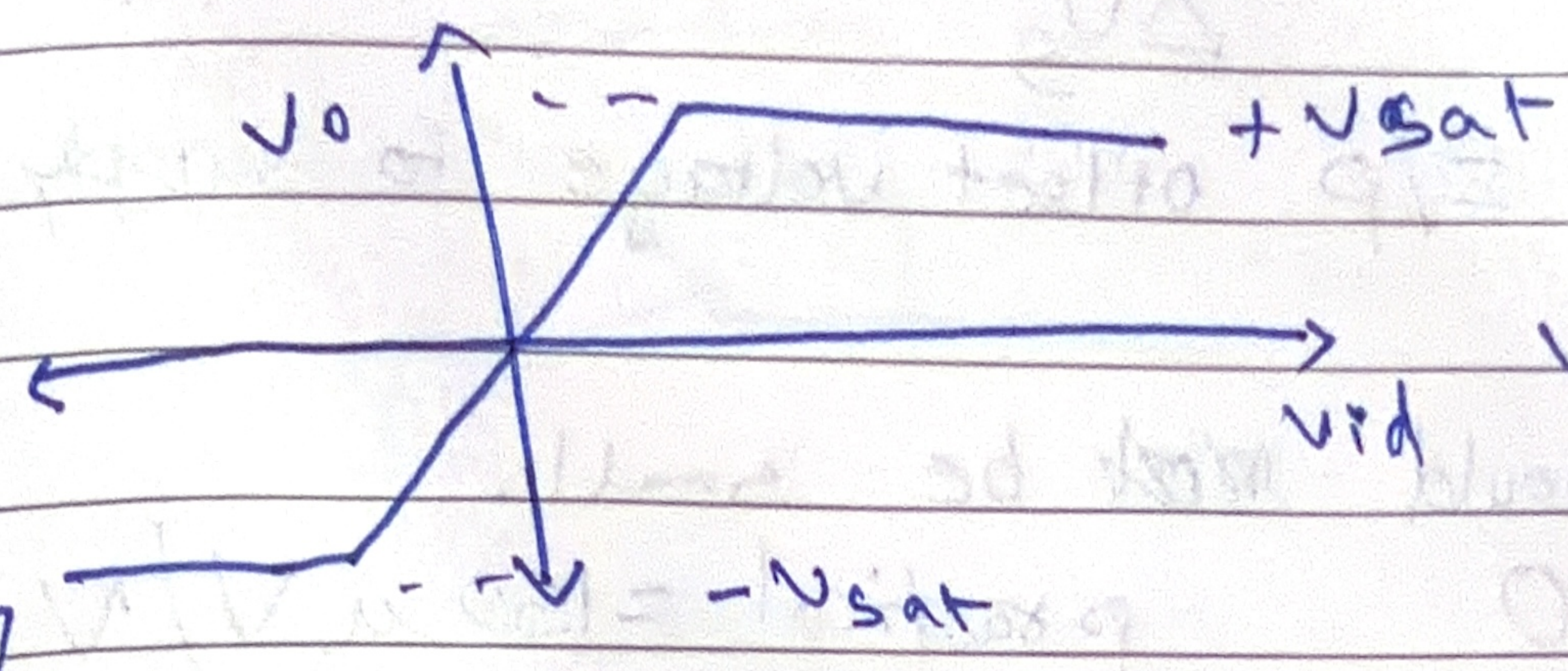
v) C

vi) Inf. slew rate

vii) 0 SVRR (Supplied Voltage rejection ratio).

Op amp works in 2 way

open loop   closed loop.



$V_{sat} = \pm 90\% \, V_{CC}$.

Op-Amp open loop, rarely used bcz of saturation

Practical features of Op-Amp

i) O/P offset voltage
voltage obtained at the o/p terminals without any I/P signal applied. denoted by (V_{OO}) ideally in mV.

ii) I/P offset voltage
def as the amount of voltage to be applied btw 2 I/P terminals such that the V_{OO} redu to zero (0). → denoted by (V_{IO}).

iii) I/P offset current
$I_{io} = |I_2 - I_1| \approx$ nano ampere

iv) I/P bias current → $I_B = \frac{I_1 + I_2}{2}$.

v) CMRR, it is the ratio of differential gain to common mode gain.
A_d / A_C → $A_d = V_o/V_{id}$ $A_C = V_o/V_i$

ideally $A_d \to \infty$ $A_c \to 0$ $CMRR \to \infty$

Practical $CMRR \approx 90dB$.

vi) large signal voltage gain, $A_d = \frac{V_o}{V_{id}}$

vii) ~~SR~~ SVRR $= \frac{\Delta V_{io}}{\Delta V_s}$. it is the ratio of change in I/P offset voltage to supply to O/P.

SVRR should ~~not~~ be small.
ideal $= 0$ practical $= 150 \mu V/V$

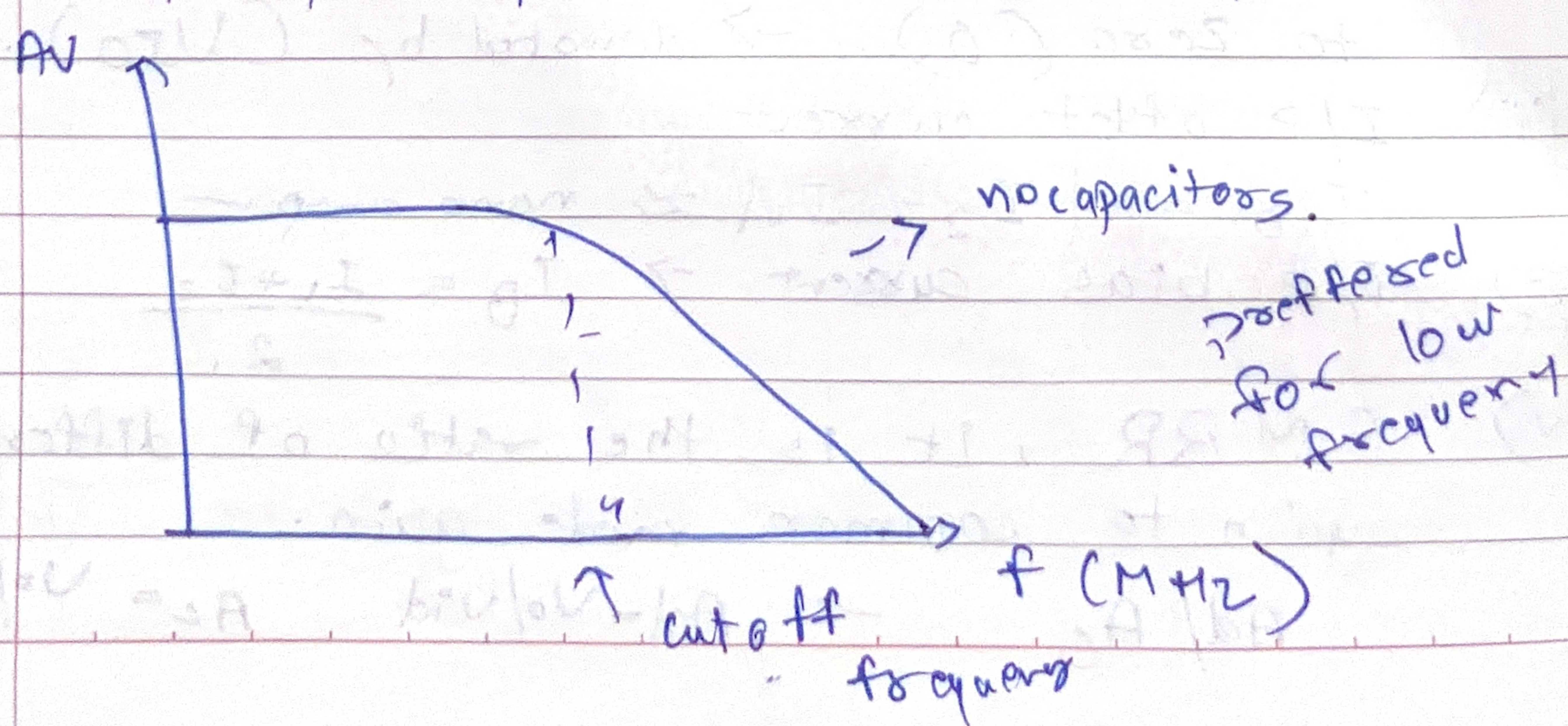
viii) Slew rate → maximum rate of change of O/P voltage per time.

$$\left.\frac{dV_o}{dt}\right|_{max} \approx 0.5 V/\mu sec \text{ (practical)}.$$

Speed of Op-Amp

ix) I/P impedance ≈ 1 MΩ
O/P " " ≈ 100 Ω [practical

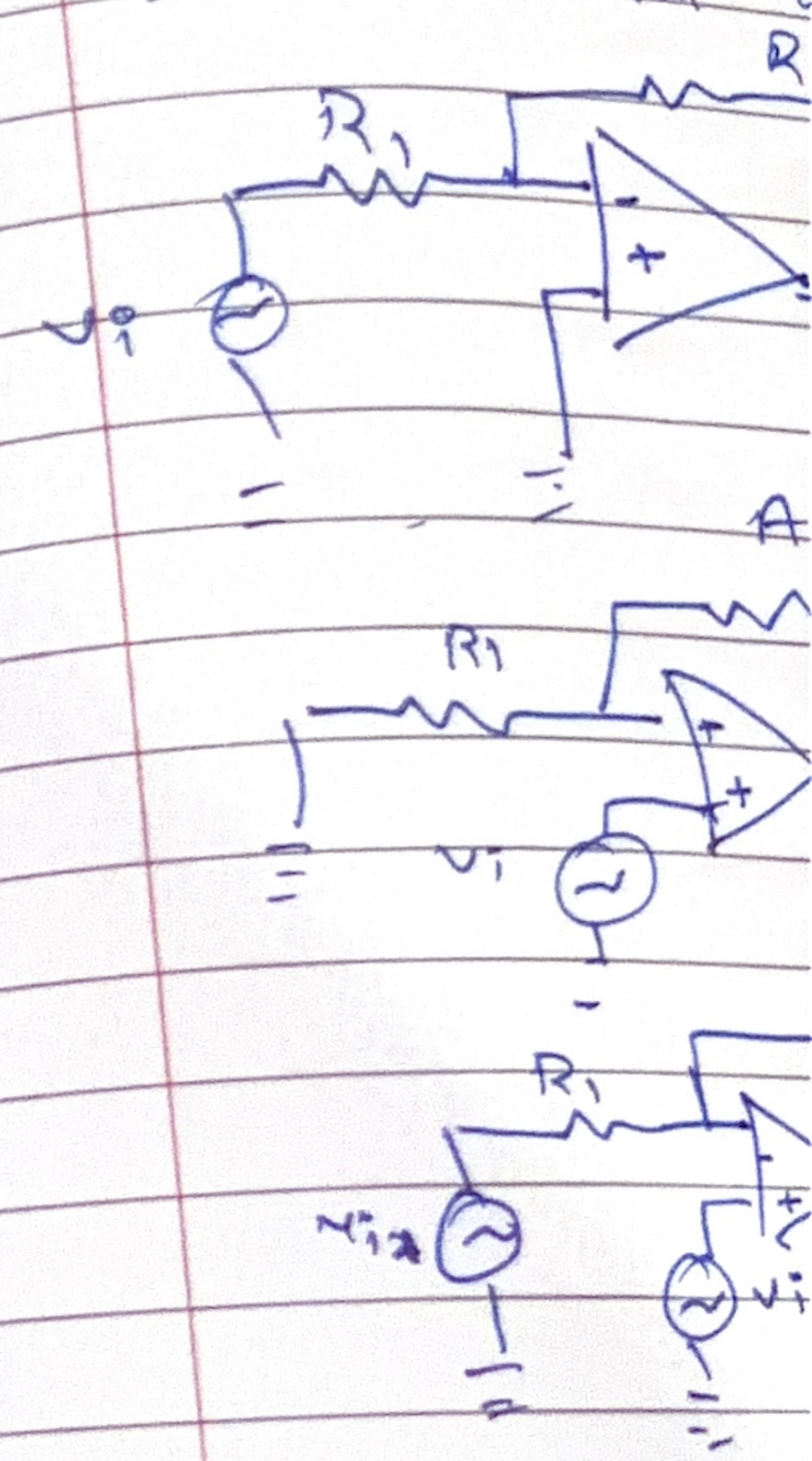
frequency response of ~~OPA amp~~ op-amp.



Also called as D...

Closed loop C...

i) inverting ampl...
ii) non-inverting am...
iii) differential ...



MRR → ∞

$\frac{V_o}{vid}$

ratio of

supply to o/p.

V/V

nge of

).

[practical]

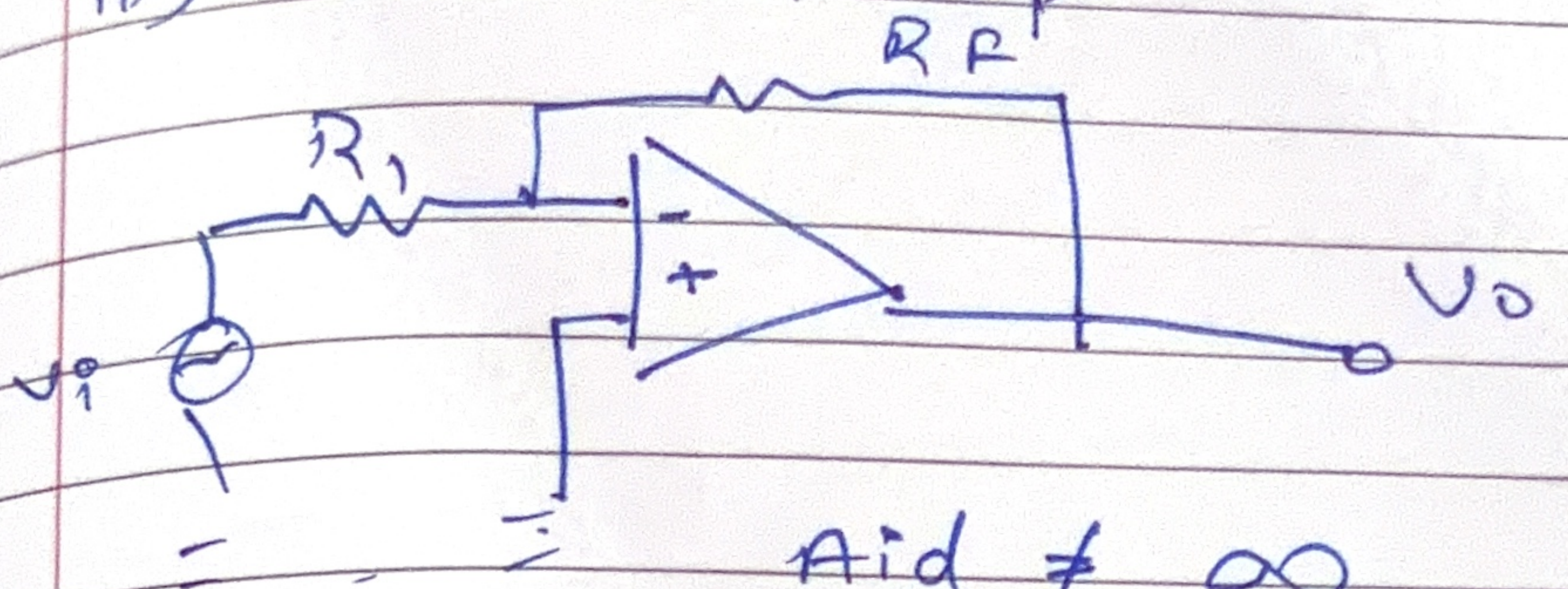
op-amp.

tered
low
requency

Also called as DC ampf.

Closed loop Op-Amp

i) inverting ampf.
ii) non-inverting ampf
iii) differential ampf.



→ inverting.

$A_{id} \neq \infty$